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First-Come,First-Served CPU Scheduling Algorithm

Write a C program to implement the First-Come, First-Served (FCFS) CPU Scheduling Algorithm. The program should take the number of processes, their Arrival Times (AT), and Burst Times (BT) as input. Calculate and display the Completion Time (CT), Turnaround Time (TAT), and Waiting Time (WT) for each process, along with the Average Turnaround Time and Average Waiting Time.

To evaluate the performance of the algorithm, we use the following formulas:

- **Completion Time (CT):** The time at which a process finishes execution.
- **Turnaround Time (TAT):** Total time taken from arrival to completion.
- $TAT = CT - AT$
- **Waiting Time (WT):** The time a process spends waiting in the ready queue.
- $WT = TAT - BT$

Input Format

1. The first input line contains an integer representing the number of processes.

Select Language : Java (OpenJDK 13.0.1)

```java
import java.util.Scanner;

public class FCFS {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n;
        System.out.println("Enter number of processes: ");
        n = sc.nextInt();

        int[] at = new int[n];
        int[] bt = new int[n];
        int[] ct = new int[n];
        int[] tat = new int[n];
        int[] wt = new int[n];

        for (int i = 0; i < n; i++) {
            System.out.println("Enter Arrival Time and Burst Time for P" + (i + 1) + ": ");
            at[i] = sc.nextInt();
            bt[i] = sc.nextInt();
        }

        int time = 0;
```

Test Cases 2

Run Sample Cases

Run All Cases

Completed

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Input Format

1. The first input line contains an integer representing the number of processes.

Select Language : Java (OpenJDK 13.0.1)

```
        int time = 0;

        for (int i = 0; i < n; i++) {

            if (time < at[i]) {
                time = at[i];
            }

            time += bt[i];
            ct[i] = time;

            tat[i] = ct[i] - at[i];
            wt[i] = tat[i] - bt[i];
        }

        double avgTAT = 0, avgWT = 0;

        for (int i = 0; i < n; i++) {
            avgTAT += tat[i];
            avgWT += wt[i];
        }

        avgTAT /= n;
        avgWT /= n;
```

Test Cases 2

Run Sample Cases

Run All Cases

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- $WT = TAT - BT$

Input Format

1. The first input line contains an integer representing the number of processes.

Select Language : Java (OpenJDK 13.0.1)

```java

            time += bt[i];
            ct[i] = time;

            tat[i] = ct[i] - at[i];
            wt[i] = tat[i] - bt[i];
        }

        double avgTAT = 0, avgWT = 0;

        for (int i = 0; i < n; i++) {
            avgTAT += tat[i];
            avgWT += wt[i];
        }

        avgTAT /= n;
        avgWT /= n;

        System.out.printf("\nAverage Turnaround Time: %.2f\n", avgTAT);
        System.out.printf("Average Waiting Time: %.2f\n", avgWT);

        sc.close();
    }
}
```

Test Cases 2

Run Sample Cases

Run All Cases

Custom Test

Success

Input :

```
3
0 2
1 2
2 3
```

Expected Output :

```
Enter number of processes:
Enter Arrival Time and Burst Time for P1:
Enter Arrival Time and Burst Time for P2:
Enter Arrival Time and Burst Time for P3:

Average Turnaround Time: 3.33
Average Waiting Time: 1.00
```

Output :

```
Enter number of processes:
Enter Arrival Time and Burst Time for P1:
Enter Arrival Time and Burst Time for P2:
Enter Arrival Time and Burst Time for P3:

Average Turnaround Time: 3.33
Average Waiting Time: 1.00
```

Sample test case

Test Case - 1 Success

Input :

```
3
0 2
1 2
2 3
```

Expected Output :

```
Enter number of processes:
Enter Arrival Time and Burst Time for P1:
Enter Arrival Time and Burst Time for P2:
Enter Arrival Time and Burst Time for P3:

Average Turnaround Time: 3.33
Average Waiting Time: 1.00
```

Output :

```
Enter number of processes:
Enter Arrival Time and Burst Time for P1:
Enter Arrival Time and Burst Time for P2:
Enter Arrival Time and Burst Time for P3:

Average Turnaround Time: 3.33
Average Waiting Time: 1.00
```

Test Case - 2 Success

Test Case - 2 Success

Input :

```
3
0 5
5 3
8 2
```

Expected Output :

```
Enter number of processes:
Enter Arrival Time and Burst Time for P1:
Enter Arrival Time and Burst Time for P2:
Enter Arrival Time and Burst Time for P3:

Average Turnaround Time: 3.33
Average Waiting Time: 0.00
```

Output :

```
Enter number of processes:
Enter Arrival Time and Burst Time for P1:
Enter Arrival Time and Burst Time for P2:
Enter Arrival Time and Burst Time for P3:

Average Turnaround Time: 3.33
Average Waiting Time: 0.00
```